

Lesson 09 Practice: Virtual Memory

JKSSB Computer Awareness

October 10, 2026

Provenance: All 20 items are original JKSSB-pattern questions written for this lesson. None is a real PYQ; each is labelled *[Original]*.

Q1. *[Original]* Virtual memory is best described as:

- A. extra physical RAM added to the computer
- B. a technique that uses secondary storage to extend the addressable memory beyond physical RAM
- C. a type of cache built into the CPU
- D. a read-only memory used to store the BIOS

Q2. *[Original]* Compared with RAM, virtual memory is:

- A. faster
- B. exactly the same speed
- C. slower
- D. always disabled

Q3. *[Original]* Which statement about virtual memory is correct?

- A. It is additional physical RAM.
- B. It is disk-backed and slower than RAM.
- C. It is faster than cache.
- D. It removes the need for RAM.

Q4. *[Original]* The main reason virtual memory exists is to:

- A. increase the clock speed of the processor
- B. allow programs and data larger than physical RAM to run, while giving each process a large contiguous view of memory
- C. replace the CPU with a faster unit
- D. store the firmware permanently

Q5. *[Original]* Expand MMU.

- A. Memory Management Unit
- B. Main Memory Utility

- C. Multiprocessor Memory Unit
- D. Machine Mapping Unit

Q6. *[Original]* The MMU is responsible for:

- A. translating virtual addresses into physical addresses
- B. storing files permanently on disk
- C. cooling the processor
- D. displaying output on the monitor

Q7. *[Original]* In paging, the fixed-size blocks into which a program's virtual address space is divided are called:

- A. segments
- B. pages
- C. frames
- D. sectors

Q8. *[Original]* In paging, the fixed-size blocks of physical memory into which pages are loaded are called:

- A. pages
- B. frames
- C. segments
- D. registers

Q9. *[Original]* Which statement correctly distinguishes paging from segmentation?

- A. Paging uses variable-size segments; segmentation uses fixed-size pages.
- B. Paging divides memory into fixed-size pages and frames; segmentation divides a program into variable-size logical segments.
- C. Both paging and segmentation divide memory into equal fixed-size blocks.
- D. Segmentation divides memory into equal frames; paging uses variable-size logical units.

Q10. *[Original]* Segmentation divides a program into:

- A. equal fixed-size pages
- B. variable-size logical segments
- C. physical frames of equal size
- D. cache lines

Q11. *[Original]* A page fault occurs when:

- A. a hardware component develops a physical fault
- B. the referenced page is not currently in RAM and must be loaded from disk
- C. the contents of RAM become corrupted
- D. the processor overheats

Q12. *[Original]* Demand paging loads a page into memory:

- A. only when it is referenced and not already present
- B. only if the whole program is loaded at startup
- C. never, because pages stay on disk
- D. only during the boot process

Q13. *[Original]* Which statement about virtual memory is **NOT** correct?

- A. Virtual memory uses secondary storage.
- B. Virtual memory is slower than RAM.
- C. Virtual memory is additional physical RAM.
- D. Virtual memory lets a process use a large contiguous address space.

Q14. *[Original]* Evaluate the statements:

- I. Virtual memory uses secondary storage to extend memory.
- II. Virtual memory is slower than RAM.
- III. Virtual memory is extra physical RAM.

- A. I and II only
- B. I only
- C. II and III only
- D. I, II, and III

Q15. *[Original]* Swapping refers to:

- A. moving data or pages between RAM and disk (secondary storage)
- B. exchanging the CPU for a GPU
- C. changing the monitor resolution
- D. moving data from cache to a CPU register

Q16. *[Original]* Thrashing is best described as:

- A. excessive paging that leaves little time for useful work
- B. a fast cache hit
- C. a type of segmentation
- D. the automatic addition of extra RAM

Q17. *[Original]* Which statement correctly distinguishes cache from virtual memory?

- A. Cache is disk-backed; virtual memory sits on the CPU chip.
- B. Cache is small, fast memory close to the CPU; virtual memory uses slower, disk-backed storage.
- C. Both cache and virtual memory are forms of secondary storage.
- D. Cache extends RAM using disk; virtual memory speeds up the CPU.

- Q18.** *[Original]* The office desktop (EX-01) has 16 GB of RAM and runs an application whose working set is larger than RAM. What does the operating system do?
- A. It refuses to run the application.
 - B. It uses virtual memory, keeping some pages on disk and loading them on demand.
 - C. It converts part of the disk into extra RAM chips.
 - D. It raises the clock speed to compensate.
- Q19.** *[Original]* From the perspective of the CPU, which of these is the slowest to access?
- A. L1 cache
 - B. RAM
 - C. disk-backed virtual memory
 - D. a CPU register
- Q20.** *[Original]* Which pair is correctly matched?
- A. Paging — fixed-size pages and frames
 - B. Segmentation — fixed-size pages and frames
 - C. Paging — variable-size logical segments
 - D. MMU — permanent file storage on disk
- Q21.** *[JKSSB FAA-pattern]* Virtual memory extends RAM by using space on the:
- A. hard disk (page/swap file).
 - B. printer.
 - C. speaker.
 - D. keyboard cable.
- Q22.** *[JKPSI-pattern]* The page file (swap file) is stored on the:
- A. hard disk/SSD.
 - B. CPU register file.
 - C. monitor screen.
 - D. mouse sensor.
- Q23.** *[SSC CGL-pattern]* Thrashing occurs when the system:
- A. spends most time swapping pages instead of executing.
 - B. prints too many pages at once.
 - C. runs only one small program.
 - D. keeps the monitor switched off.
- Q24.** *[JKSSB-pattern]* (MSQ: select **ALL** correct options.) Which of the following relate to virtual memory?
- A. Paging

- B. Swap/page file
 - C. Hard-disk space used as memory extension
 - D. Printer spooler
- Q25.** *[SSC CGL-pattern]* **(MSQ: select ALL correct options.)** Which statements about virtual memory are correct?
- A. Paging moves fixed-size blocks between RAM and disk.
 - B. A page fault triggers loading the needed page from disk.
 - C. Thrashing means excessive paging with little progress.
 - D. Virtual memory is a special RAM chip on the motherboard.
- Q26.** *[PYQ-adapted: JKSSB Junior Assistant 2026 Q62, final key; SRC-03/SRC-06]* Virtual memory uses part of which resource to extend the apparent capacity of main memory?
- A. Secondary storage
 - B. CPU register file
 - C. Printer buffer
 - D. ROM firmware area
- Q27.** *[Original]* A page currently used by a process is absent from RAM but present in its backing store. The operating system typically responds with a:
- A. Page fault
 - B. Cache hit
 - C. Power failure
 - D. Print spool
- Q28.** *[Original]* Virtual memory allows a process to address a space that can be:
- A. Larger than the available physical RAM
 - B. Stored only in CPU registers
 - C. Volatile only when the disk is disconnected
 - D. Equal to the size of the cache
- Q29.** *[Original]* Which statement best distinguishes virtual memory from cache?
- A. Virtual memory extends the apparent main-memory space; cache speeds up frequent access.
 - B. Both are permanent backups.
 - C. Cache replaces the operating system.
 - D. Virtual memory is a CPU register.
- Q30.** *[Original, MSQ drill]* **(MSQ: select ALL correct options.)** Which statements about virtual memory are correct?
- A. It is managed with operating-system support.
 - B. A page fault can occur when a needed page is not in RAM.
 - C. Backing storage may be used for non-resident pages.
 - D. It makes secondary storage as fast as registers.